

# Review 1:

This paper introduces and formalizes "value profiles", an architecture for context-dependent behavioural control within the active inference framework. The central goal is to achieve flexible, state-conditional behaviours without the computational burden of maintaining independent value parameters—specifically, outcome preferences, policy priors, and policy precision—by using a small library of reusable parameter "bundles" that are dynamically recruited via belief-weighted mixing as the agent accomplishes the task. Through simulation-based model and parameter recovery, the paper aims to show that this approach can be applied to simple datasets.

## Major Comments:

The approach presented in the manuscript has **several key flaws and limitations**.

Primarily, the methodological choices raise concerns about the **practical applicability of the approach and its usefulness**. They assume substantial prior knowledge about the Z definition, including the number of behavioural modes needed and the mapping from contexts to profiles, while treating the generative model parameters A, B, and D as fixed and known. But more surprisingly, the authors treat  $\gamma$  as a bundle parameter, claiming that its value steers the selection of exploratory/exploitative policies. In AIF,  $\gamma$  is adaptively computed by considering the distribution of the policy qualities computed through the expected free energy. Its role is to augment/reduce the stochasticity of policy selection, and **it does not influence the behavioural properties of the policies**.

Another significant limitation is the **gap between the model, the experimental setting, and real-world complexity**. The reversal learning task is highly simplified because it involves only two contexts with fixed transitions. Real-world environments often involve gradual drift, stochastic regime switches, ambiguous boundaries, or an unknown number of latent contexts that must be discovered.

An additional concern is the **absence of an experimental comparison between the proposed profiles recruitment and both a hierarchical approach and a more straightforward AIF method with learning**. The authors, in fact, claim in Section 1.5 that the profiles are an "amortized form of hierarchical modulation", but this amortization could come at the cost of some compositional flexibility, in the sense that hierarchical active inference could combine multiple levels and timescales in ways the profile architecture cannot (see, for instance, "Active inference and cognitive control: Balancing deliberation and habits through precision optimization", of Proietti et al, Physics of life reviews, 2025, Elsevier).

Mutatis mutandis for AIF with learning: knowing a priori the number of potential contexts, it is possible to let AIF learn the parameters by detecting differences between the

generative model and the generative process, in a way that is much more tractable and scalable than the exhaustive two-stage grid search implemented in the proposed approach.

### **Minor Comments:**

There are a few minor formatting and editing issues.

In Section 2.7.3, Profile 0 is called “explorative” while Profile 1 is called “exploitative”, where  $E$  encodes policy-prior logits. But using the softmax function on the two  $E$  vectors, we find that the distribution of Profile 0 is more peaked than that of Profile 1.

In the text,  $E$  seems to have the cardinality of the action set. Although not explicitly specified, I deduce that the planning horizon is 1.

I could not have found where Fig. 2 is mentioned in the text.

## **Review 2:**

This paper introduces value profiles, a set of predefined reusable parameter bundles for goal priors, policy priors and action precisions. Such profiles are selected (either through one-hot encoding or via soft belief mixing) based on a context state factor inferred from observations. This mechanism permits efficiently switching between a compact set of behaviors (e.g., exploratory vs goal-seeking), which remain stable regardless of subtle variations of the environment. Compared to hierarchical models, the proposed model could perform better in previously learned tasks, as different behavioral modalities could be mapped at a low level for fast execution, without requiring slow learning or higher-level planning. Another advantage compared to standard models is the possibility of dynamically changing habits and goals, since the latter are usually fixed while the former are slowly learned. The paper is clearly written, the proposal is sound and supported by the simulation results. I believe it is well suited for Biological Cybernetics.

Here are a few suggestions to improve the paper:

- In Section 2.3, it might be better to explicitly define every hidden state factor (context, better\_arm, etc.) and outcome modalities ( $o=[o\_hint, o\_reward, o\_choice]$ ), including how many states they can assume (e.g., null, loss, or reward for the reward observation, etc.). This could also be done in a visual representation of the two armed bandit task (similar to Smith et al., Figure 4). Also, are  $q(s\_ctx)$  and  $q\_context$  the same posterior? Importantly, it would be very informative for the readers to add a graphical representation of the generative model, specifying the likelihood matrices (e.g., the likelihood matrix associated with the context factor only predicts reward observations), and highlighting the inference over the assignment state factor in addition to standard state inference, as

well as the dependency of habits, goals, and precision from the set of reusable parameter bundles.

- The author could expand on inference and control for standard active inference models (Section 2.4) in an additional section in the Supplementary Material, for those unfamiliar with the active inference framework. Specifically, describing how the policy priors at one level are predicted by a higher level in hierarchical active inference (see, e.g., Friston et al., 2025, From pixels to planning: scale-free active inference, Figures 1 and 2) could help highlight the differences with the proposed model, in which policy priors are selected by the context state factor instead.

- In section 2.7.1, if  $C$  denotes preferences over reward observations only, perhaps it would be better to name the vector  $C_{\text{reward}}$ , to distinguish it with the preferences of the other two observation modalities (hints and choices).

- One aspect that is never discussed is how the entropy of the policy posterior changes across models (in particular M1 and M2) as a function of the EFE and the uncertainty of the better\_arm state factor. In addition to just plotting the precisions in Fig. 1C, it could be useful to plot the policy entropy of M1 and that of M2 (which is also influenced by the dynamic precision  $\gamma_t$ ) to see how they differ. If M1 responds to uncertainty (not through  $\gamma$  directly, but through the entropy of the policy posterior), the key limitations of M1/M2 in Table 1 should be changed accordingly.

- Section 3.3.3: it would be valuable to see how model M3 behaves in a simple simulation in which the actual context changes from volatile to stable at mid trial, plotting the dynamics of the precision  $\gamma$  and  $q_{\text{context}}$ . This is only briefly mentioned in Section 4.2.1 among other things, but it could also be addressed in a future work.

- Besides the learning of  $Z$ , one very interesting aspect I would like to see as future work is the possibility of having a different set of outcome preferences based on the inferred context; this is only proposed in theory and not implemented in practice. Inferring dynamic goals from sensations is a behavior frequently observed in biological organisms that has not yet been analyzed in detail in the active inference literature. To this matter, one aspect to analyze regards conflict resolution and the Buridan's ass: in the presented work, the set of parameter bundles used allows for smooth mixing of exploration and exploitation behaviors. Considering a set of opposing goal-directed behaviors, if context inference is unreliable and the weights are not sharply peaked, could the agent be stuck with conflicting outcome preferences? Would increasing the precision of the assignment matrix be a possible solution? Also, regarding hierarchical settings: the model assumes that policy priors and context solely rely on outcomes. What happens if we let habits and goals be shaped both by observations and high-level priors? This would allow for the investigation of different scenarios (e.g., mixing fast but reactive policies based on current sensations with slower high-level planning) and degenerate conditions (e.g., depending on the ratio between high-level and low-level precisions, the agent could infer

a stable context despite evidence of environmental volatility). The author could mention these aspects in an additional brief Section after 4.2.3.

- The following references may also be relevant: (Proietti et al., 2025, "Active inference and cognitive control: Balancing deliberation and habits through precision optimization") about adaptable policy precisions based on context; (Priorelli et al., 2025, "Embodied decisions as active inference") about cautious/impulsive behaviors and inference of decisions and goals from continuous observations; (Van de Maele et al., 2024, "A hierarchical active inference model of spatial alternation tasks and the hippocampal-prefrontal circuit") about hierarchical message passing between goals and policies.

Overall, this is a well-written paper that introduces a useful mechanism for adaptive behavior in active inference, and the simulations support the main claims. I therefore recommend publication after minor revisions.